MILITARY SPECIFICATION

ADAPTER, LAUNCHER, AIRCRAFT, ADU-299A/A

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification establishes the requirements for one type of launcher adapter that can be suspended from various ejector bomb racks and permits a guided missile launcher to be mounted to its bottom surface.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-T-7743	Testing, Store Suspension Equipment, General Specification For

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

FSC 1055

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STANDARDS (Continued)

Military

MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3314	Lug, Suspension, (1000 LB Class), Airborne Equipment

DRAWINGS

Naval Air Systems Command

DL564AS100	Adapter Assembly ADU-299A/A Aircraft Launcher
54A46C358	Lug, Suspension, Airborne Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article - The adapter furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.3).

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to applicable specifications or standards and shall be as specified herein and on applicable drawings.

3.4 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

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3.5 Design and construction - The design and construction of the adapter shall be as specified herein and in accordance with drawings listed in DL564AS100.

3.6 Performance - The adapter shall be capable of being suspended from any of the AERO 7 series, MAU 9 series, BRU-10 and 11 series, Aero 20A series and ERU-119 ejector type bomb racks using either the 14 inch or 30 inch suspension lugs. Mounted on any of these ejector bomb racks the adapter shall accept either the LAU-7/A series or AERO 3A series guided missile launcher for mounting to its bottom surface. Interconnection of adapter electrical cables and launcher electrical cables shall be made manually. The adapter shall satisfy all performance requirements when subjected to the following tests:

(a)	Examination	(4.6.1)
(b)	Electrical Continuity	(4.6.2)
(c)	Operational Tests	(4.6.3)
(d)	Shock Tests	(4.6.4)
(e)	Electrical Characteristics	(4.6.5)
(f)	Static Tests	(4.6.6)
(g)	Vibration Test	(4.6.7)
(h)	Salt Spray Test	(4.6.8)

3.7 Interchangeability - All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in part number shall be governed by the drawing number requirements of MIL-STD-100.

3.8 Identification of parts - All parts used in the adapter shall be marked with the manufacturer's code number in accordance with MIL-STD-130.

3.9 Workmanship - The workmanship displayed in fabrication and assembly of the adapter shall be such as to assure, within design limitations, ability of the adapter to meet performance requirements under all applicable environmental conditions specified herein. Unauthorized repair, welding, heavy burrs, or parts assembled by introduction of high stresses not prescribed in design, are typical signs of inferior workmanship and shall be cause for rejection. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship. All electrical components and wiring shall be installed in accordance with the workmanship requirements of MIL-E-5400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure the supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of the adapter shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)

4.3 First article inspection - First article inspection shall consist of all of the inspections listed in Table I. First article inspection report format shall be in accordance with MIL-STD-831 (see 6.4).

4.3.1 Sampling for first article inspection - First article adapter sample shall consist of the first two adapters manufactured on contract or order. The adapters shall be labeled A and B for ease of identification. Failure of the adapter assembly or any subassembly to pass any of the inspections shall be cause for rejection.

TABLE I
FIRST ARTICLE INSPECTIONS - SCHEDULE AND SEQUENCE 1/

TESTS	PARA. NO.	SAMPLE	
		A	B
Examination, adapter	4.6.1	1	1
Electrical continuity test	4.6.2	2	2
Operational test	4.6.3	3	3
Shock test 2/	4.6.4	x	4
Electrical characteristics	4.6.5	4	x
Static test 2/	4.6.6	x	6
Vibration test 2/	4.6.7	x	5
Salt spray 2/	4.6.8	5	x

1/ Test sequence is indicated by the numbers in the sample column. An x indicates test not required.
2/ Repeat the electrical continuity test after this test (see 4.6.2).

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4.4 Quality conformance inspection - Quality conformance inspection shall consist of the following inspections:

a. Individual inspections (4.4.1)
b. Sampling inspections (4.4.2)

Quality conformance inspection report format shall be in accordance with MIL-STD-831 (see 6.4).

4.4.1 Individual inspections - Individual inspections are those inspections conducted on each adapter in accordance with the inspections listed in Table II, Group 1. Failure to pass any one of these inspections shall be cause for rejection of the entire lot.

4.4.2 Sampling inspections - A random sampling shall be selected by the procuring activity from each production lot in accordance with Table III and shall be subjected to the inspections listed in Table II, Group 1 and Group 2. Group 1 inspections shall be conducted prior to Group 2 inspections. Failure of any sample to pass any one of these inspections shall be cause for rejection of the entire lot.

TABLE II
QUALITY CONFORMANCE INSPECTION AND SEQUENCE 1/

TESTS	PARA NO.	SAMPLE			
		A	B	C	D
Group 1 - Individual Tests					
Examination, adapter	4.6.1	1	1	1	1
Electrical continuity	4.6.2	2	2	2	2
Group 2 - Lot Sample Tests					
Operational tests	4.6.3	3	3	3	3
Shock tests 2/	4.6.4	4	x	4	x
Electrical characteristics	4.6.5	x	4	x	4
Static tests 2/	4.6.6	6	x	5	x
Vibration tests 2/	4.6.7	5	x	x	x
Salt spray 2/	4.6.8	x	5	6	x

1/ Test sequence is indicated by the numbers in the sample column. An x indicates test not required.

2/ Repeat the electrical continuity test after this test (see 4.6.2).

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TABLE III

SAMPLING INSPECTION

ITEM	PRODUCTION LOT SIZE	NO. OF SAMPLES TO BE TESTED
Adapter	100 or less 101 to 200 201 to 300	2 3 4

4.5 Inspection conditions - Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Room temperature - room ambient of 25° ±10°C (77° ±18°F).

b. Any ejector bomb rack specified in paragraph 3.6 and either the LAU-7/A series or AERO-3A series launcher shall be used for mounting and loading the ADU-299A/A adapter for any of the required tests. The 14 inch suspension lugs shall be used to attach the adapter to the bomb rack. During all applicable tests the bomb rack sway braces shall be adjusted such that they impose not more than 500 pounds per brace direct loading on the adapter.

c. Unless otherwise specified a sidewinder 1C (AIM-9D) simulated missile shall be used as the test store. A simulated configuration having the same center of gravity location and sidewinder 1C type mounting lugs can be used as an alternate test store.

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4.6 Test methods -

4.6.1 Examination, adapter - Each adapter shall be thoroughly examined to determine conformance to all of the requirements of this specification and applicable documents.

4.6.1.1 The adapters shall be examined to ascertain that the preparation for delivery conforms to this specification.

4.6.2 Electrical continuity - Electrical circuit continuity of the ADU-299A/A adapter shall be checked in accordance with the applicable wiring and electrical schematic drawings 564AS103 and 67A71C8. After the adapter is completely assembled all electrical cable wiring shall be tested for continuity. Continuity failure shall be cause for rejection.

4.6.3 Operational tests -

4.6.3.1 Bomb rack mounting - Install two (2) 14 inch bomb suspension lugs (MS3314) into the proper adapter mounting holes (see dwg 564AS100). Verify that the adapter shall latch securely on the 14 inch suspension hooks of an applicable ejector bomb rack (see 3.6). Adjust bomb rack sway braces in accordance with 4.5b. Repeat test removing 14 inch suspension lugs and installing two (2) 30 inch bomb suspension lugs (54A46C358) into the proper adapter mounting holes.

4.6.3.2 Launcher mounting - A LAU-7A or AERO-3A launcher shall be firmly mounted (165-200 ft lbs torque) against the bottom mounting pads of the ADU-299A/A adapter. This mounting shall be accomplished by using the 30 inch spaced launcher mounting bolts which are tightened into the 30 inch spaced adapter nuts (see dwg 67A71B5). During attachment of the launcher to the adapter, electrical cables shall be manually interconnected with their corresponding mating connectors.

4.6.4 Shock tests - This test shall be conducted in accordance with the shock test of MIL-T-7743 except for changes specified herein. The shock tests specified in MIL-T-7743 shall be modified where necessary so that the tests are accomplished in each of the six directions parallel to the three principal axis of the adapter. The control accelerometers shall be located at the AERO-7A bomb rack/ADU-299A/A adapter interface. The exact location shall be on the bomb rack adjacent to each adapter lug. With the adapter mounted on an AERO 7A bomb rack (14" suspension) and supporting a LAU-7/A or AERO-3A launcher, and a sidewinder 1C missile, or equivalent, the load factors ("G" values) shall be in accordance with Table IV. Yield shock loads are 1.15 times the values shown in Table IV.

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TABLE IV

TABLE OF LIMIT SHOCK LOADS (G's)

DIRECTION	LOAD FACTORS (G's)
Vertical up	7.1
Vertical down	12.5
Longitudinal forward	9.8
Longitudinal aft	9.8
Transverse port	5.2
Transverse starboard	5.2

4.6.4.1 Only the following shock tests specified in MIL-T-7743 are to be conducted on the adapter:

a. No load condition - Shocked once in each of the six (6) directions to 25 ±2.5G for 30-100 milliseconds.

b. Yield load condition - Shocked twice in each of the six (6) directions as per Table IV for 30-100 milliseconds.

4.6.5 Electrical characteristic - Insulation resistance tests shall be performed on the adapter in accordance with MIL-T-7743.

4.6.6 Static tests - Static load tests shall be conducted in accordance with the static tests of MIL-T-7743 except for the changes specified herein. Table V contains the loads that shall be used for the test where applicable. The loads shall be applied at the center of gravity of the launcher/missile combination (see para. 4.5). The LAU-7A launcher/missile combination shall be mounted to the adapter as per 4.6.3.2. The adapter/launcher missile combination shall be mounted using 14" suspension on the AERO-7A bomb rack. Permanent deformation, cracks or rupture shall be cause for rejection.

4.6.6.1 Test conditions -

a. Loads A, B, and C applied as specified in Table V.

b. Loads A and C applied as specified in Table V and with load B applied in the aft direction.

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TABLE V

STATIC LOAD TESTS

LOAD	LOAD TESTS
A	2400 pound (ultimate) applied in a direction parallel to the launcher mounting bolts and with the missile below the launcher in a downward direction.
B	3000 pound (ultimate) applied in a forward direction along the longitudinal axis of the missile.
C	6000 pound (ultimate) applied perpendicular to loads A and B.

4.6.7 Vibration test, resonance - This test shall be conducted in accordance with the resonance vibration tests of MIL-T-7743 except for changes specified herein. The vibration testing shall consist of only the resonance portion of the tests specified in MIL-T-7743. Minimum time period of vibration for each resonance search shall be 20 minutes unloaded and 20 minutes loaded. Release and transfer at resonance vibration is not applicable to this adapter except that the 5 minute period of vibration at each resonant point as specified in MIL-T-7743 shall be accomplished. If no resonance or complex motion occurs the test configuration shall be vibrated at 50 cps. All vibration frequencies shall be $\pm$2% and amplitudes $\pm$5%. The basic bomb rack/adapter/launcher/missile test configuration shall be as specified in paragraph 4.5 and 4.6.6 except as modified by test criteria.

4.6.8 Salt spray - This test shall be conducted in accordance with the salt spray test of MIL-T-7743 except for changes specified herein. After exposure of the adapter to the basic salt spray for the 50 hour and the 48 hour drying period the adapter shall be visually examined and tested for electrical continuity (see 4.6.2).

5. PACKAGING

5.1 Preservation and packaging - The adapter shall be preserved and packaged in accordance with MIL-STD-794, level B.

5.2 Packing - The adapter shall be packed in accordance with MIL-STD-794, level B.

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5.3 Marking and labeling -

5.3.1 Unit packages - Each unit package shall be durably and legibly marked or labeled with the following information and in accordance with the requirements of MIL-STD-129.

Adapter, Launcher, Aircraft, ADU-299A/A
Military Specification No. MIL-A-81918(AS)
Preserved (Date)
Domestic or Export Packed (as applicable)
Government Order No. (or Contract No. if order no. is not assigned)
Names of Supplier (and name of Manufacturer if not the same)

5.3.2 Marking - All shipping containers shall be marked in accordance with applicable requirements of MIL-STD-129.

6. NOTES

6.1 Intended use - The adapter established by this specification is intended to be suspended from aircraft ejector bomb racks to permit the mounting of LAU-7/A or AERO-3A missile launchers at the same bomb rack stations.

6.2 Information for contracting officer - Contracts or orders shall specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing required (see 5.1 and 5.2).

c. Samples subjected to sampling inspection shall not be considered or accepted as part of the contract or order (see 4.4.2).

d. Determine sampling test selection (see 4.4.2).

e. Items of data required (see 6.4).

f. Name and location of approved Government Test Laboratory.

6.2.1 Contract or orders shall specify the following provision for first article inspection.

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6.2.1.1 First article - When a first article is required for inspection and approval (see 3.1, 4.3, 6.2 and 6.4). When a contractor is in continuous production of adapters from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

(a) If first article tests are conducted at the contractor's plant or a commercial laboratory, a test report shall be forwarded for verification to the procuring activity.

(b) That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Availability of equipment for tests - The contractor may submit a request to the contracting officer for an ejector bomb rack, a LAU-7/A or AERO-3A launcher to be used for the tests specified herein.

6.4 Contract data requirements - When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Reports	DI-T-5329 - Inspection Test Reports
4.4	Quality Conformance Inspection Reports	DI-T-5329 - Inspection Test Reports

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.6 Quality assurance definitions - Definitions for quality assurance terms are in accordance with MIL-STD-109.

Preparing activity:
Navy - AS
DOD Proj. No. 1055-N053

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OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

MIL-A-81918(AS) ADAPTER, LAUNCHER, AIRCRAFT, ADU-299A/A

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO (If "Yes", in what way?)

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